



SYSTEM INTEGRATION AND ARCHITECTURE

MONTHLY PROGRESS REPORT

Student Information

DATE SUBMITTED	July 22, 2026	MPR NO:	5
SECTION	3A		
PROJECT TITLE	Development of a Web-Based Resident Management and Service Automation System for Barangay Fortune		
STUDENTS NAME	<i>LAST NAME, FIRST NAME MI. – DESIGNATION</i> 1. Adona, Andrei Joshua A. - Communication Leader 2. Bebar, Ivan Mark R. - Document Leader 3. De Dios, Owen Hart T. - Technical Leader 4. Magpayo, Aironn Paul M. - Project Leader 5. Siupan, Ivan S. - Quality Assurance Leader		

1. Tasks Planned for the Month

(What the group intended to accomplish)

Develop a comprehensive E-Serbisyo Portal encompassing distinct workflows for Residents, Front Desk Staff, and Administrators.

Implement a robust Official Receipt (OR) system to track payments and finalize document releases securely.

Implement a Daily Appointment Limit (Queue Management) to prevent staff overload on specific dates.

Develop strict role-based access control with granular privileges for staff members.

Implement an automated Audit Log system to track sensitive administrative and financial actions.

Set up automated email notifications for document statuses and password recovery.

Optimize database performance and prevent server storage bloat.

2. Tasks Accomplished

(What was completed)

Official Receipt (OR) System: Developed dynamic OR code generation.

Integrated a digital Receipt Preview modal and an OR submission flow, allowing staff to officially record transactions into the Audit Logs when documents are released to residents.



Queue & Daily Limits: Engineered a dynamic capacity-checking system. The Resident's appointment calendar automatically disables dates that have reached the maximum daily queue limit, preventing overbooking.

Role-Based Dashboards & Privileges: Created secure, isolated dashboards for Residents, Staff, and Admins. Added granular controls allowing Admins to grant or revoke specific "Pending Review" privileges to individual staff members.

System Audit Logs: Built a comprehensive audits_logstable that meticulously tracks critical actions such as OR submissions, account suspensions, captain assignments, and password resets, complete with exact timestamps and user references.

Email Integration & Unified Recovery: Integrated Nodemailer to automatically dispatch email updates (e.g., Approval, Ready for Pickup). Built a unified and secure password reset flow covering both Residents and Staff/Admins.

Dynamic UI/UX Optimizations: Implemented real-time 5-second asynchronous polling for live notifications. Replaced hardcoded elements by introducing dynamic document attachment toggling and dynamic Barangay Captain signature assignments for printed certificates.

Storage Optimization: Engineered automatic garbage collection using fs.unlink() to instantly delete physical file uploads (IDs, proofs) when requests are cancelled by residents or rejected by staff.

3. Current System Status

(Brief Description of Progress)

Health: The system is highly stable and fully operational for end-to-end document requesting, reviewing, printing, and releasing.

Database: Fully structured with normalized relational tables handling Residents, Officials, Document Templates, Queue Management, Password Resets, and Audit Logs.

Performance: The implementation of lightweight asynchronous polling provides a "real-time" app experience without overloading the server. Server disk space is highly optimized due to the automated file cleanup routines.

4. Problems Encountered

(If none, write "None")

Problems Encountered:

- Generating a unique, verifiable OR code that both the resident and the staff could reference securely during the physical claiming process.
- Preventing users from overbooking a specific date required complex, real-time database calculations before rendering the React dropdown calendar.
- Leftover physical files remained on the server indefinitely when requests were cancelled or rejected, leading to severe potential storage bloat.



- Hardcoded values (such as the Barangay Captain's name on certificates or document attachment requirements) made the system rigid and required manual code intervention to change.
- Managing state and security routing across three entirely different user roles (Admin, Staff, Resident) while preventing unauthorized access to specific dashboard components.

5. Actions Taken / Solutions

Created a unique hashing algorithm (generateORCode) based on the primary request_id and a mathematical salt to ensure the OR code is distinct, deterministic, and verifiable.

Wrote a specialized backend endpoint (/available-dates) that aggregates existing requests per day, flags them as isFull if they hit the daily threshold, and seamlessly disables those options in the frontend UI.

Implemented robust fs.unlink() commands in the backend rejection/cancellation API routes to immediately and permanently delete associated files when a request is dropped.

Migrated rigid hardcoded values into dynamic database flags (is_captain, requires_attachment) and constructed specific UI toggles for Admins to control them dynamically without developer assistance.

Implemented strict JWT/localStorage verification checks and created specialized layout components (e.g., AdminSidebar, ResidentBottomNav) to isolate role environments securely.

Prepared by:

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(Name over Signature)

Date: July 22, 2026

Checked By:

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